

MECH5200 Numerical Methods for PDEs : *Video 21:* *The Weak Form*

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Weak Form

- Finite elements uses the **weak form** of the governing equation.
- Consider a PDE, represented using the functional $\mathcal{L}(u)$:

$$\mathcal{L}(u) = f \quad (1)$$

- With Dirichlet or Neumann boundary conditions:

$$u_{\Gamma_D} = g \quad (2)$$

and:

$$\frac{\partial u}{\partial n}|_{\Gamma_N} = h \quad (3)$$

Weak Form

- What if we multiply this by a function w and integrate across the domain of interest:

$$\int_{x_L}^{x_R} w \mathcal{L}(u) dx = \int_{x_L}^{x_R} w f dx \quad (4)$$

- Solving this expression is effectively the same as solving the original equation.
 - **BUT** what does it mean to do this?
 - Let's look at what happens when w is a linear hat basis function.

Weak Form

- Graphically, it shows that we are satisfying a weighted expression of the equation in the domain/sub-domain:

$$\mathcal{L}(\text{red } u) = \text{green } f$$

$$\mathcal{L}(\text{red } u) - \text{green } f = \text{blue } R$$

$$\int \text{purple } w \cdot \mathcal{L}(\text{red } u) dx - \int \text{purple } w \cdot \text{green } f dx = \int \text{purple } w \cdot \text{blue } R dx = 0$$

$$\boxed{\int \text{purple } w \cdot \mathcal{L}(\text{red } u) dx - \int \text{purple } w \cdot \text{green } f dx = \int \text{purple } w \cdot \text{blue } R dx = 0}$$

Poisson's Equation?

- Let's do an example with the Poisson equation/Laplace operator in one dimension:

$$\frac{\partial^2 u}{\partial x^2} = f \quad (5)$$

- Let's re-arrange the equation a little:

$$\frac{\partial^2 u}{\partial x^2} - f = R = 0 \quad (6)$$

- This shows that the equation has a *zero residual*, R when solved exactly.

Weak Form

- Let's now multiply the expression by w , a weighting function and integrate across the domain/region of interest:

$$\int_{x_L}^{x_R} \left(w \frac{\partial^2 u}{\partial x^2} \right) dx - \int_{x_L}^{x_R} (w f) dx = \int_{x_L}^{x_R} w R dx = 0 \quad (7)$$

- For this weighting function, the result is an integral expression over the whole domain.
- Note: most of the function w is equal to zero (i.e., local basis function).
- Notice, that the *residual* has been weighted and integrated – and remains zero.

The weak form

- Let's now integrate this **weighted residual** integral expression by parts:

$$\int_{x_L}^{x_R} \left(w \frac{\partial^2 u}{\partial x^2} \right) dx - \int_{x_L}^{x_R} (wf) dx = 0 \quad (8)$$

- Recall, integration by parts in one dimension:

$$\int_{x_L}^{x_R} f(x) g'(x) dx = f(x) g(x) \Big|_{x_L}^{x_R} - \int_{x_L}^{x_R} f'(x) g(x) dx \quad (9)$$

- Let's set $f(x) = w$ and $g'(x) = \frac{\partial^2 u}{\partial x^2}$, then:

$$\int_{x_L}^{x_R} \left(w \frac{\partial^2 u}{\partial x^2} \right) dx = w \frac{\partial u}{\partial x} \Big|_{x_L}^{x_R} - \int_{x_L}^{x_R} \frac{\partial w}{\partial x} \frac{\partial u}{\partial x} dx \quad (10)$$

The weak form

- Inserting the result from equation 10, into equation 8:

$$\underbrace{w \frac{\partial u}{\partial x} \Big|_{x_L}^{x_R} - \int_{x_L}^{x_R} \frac{\partial w}{\partial x} \frac{\partial u}{\partial x} dx - \int_{x_L}^{x_R} (w f) dx}_{\int_{x_L}^{x_R} \left(w \frac{\partial^2 u}{\partial x^2} \right) dx} = 0$$

- Let's focus only on the reformulated expression:

$$w \frac{\partial u}{\partial x} \Big|_{x_L}^{x_R} - \int_{x_L}^{x_R} \frac{\partial w}{\partial x} \frac{\partial u}{\partial x} dx - \int_{x_L}^{x_R} (w f) dx = 0 \quad (11)$$

- Which we will call the **weak form**. Embedded in this equation is the assumption that the underlying function is continuous (no discontinuities).

The weak form

- The final weak form of the PDE is (for weighting function w_i):

$$\underbrace{w_i \frac{\partial u}{\partial x} \Big|_{x_L}^{x_R}}_{\text{at boundaries}} - \underbrace{\int_{x_L}^{x_R} \frac{\partial w_i}{\partial x} \frac{\partial u}{\partial x} dx}_{a(u, w_i)} - \underbrace{\int_{x_L}^{x_R} w_i f dx}_{f(w_i)} = 0 \quad (12)$$

- $a(w, u)$: Is the bilinear form
- Note: If we associate w_i with each node in the domain, we have defined i equations.
- Often this is written as:

$$a(u, w) - f(w) = 0 \quad (13)$$

- This final weak form contains integrals of first derivatives, whereas in the strong form there was a second derivative.

What have we learned

- Let's look at the graphic again:

$$\mathcal{L}(\text{red } u) = \text{green } f$$

$$\mathcal{L}(\text{red } u) - \text{green } f = \text{blue } R$$

$$\int \text{purple } W \cdot \mathcal{L}(\text{red } u) dx - \int \text{purple } W \cdot \text{green } f dx = \int \text{purple } W \cdot \text{blue } R dx = 0$$

$$\boxed{\int \text{red } u \cdot \text{green } f dx - \int \text{green } f \cdot \text{red } u dx = \int \text{blue } R dx = 0}$$

- Integration by parts results in a reduction of order of PDE.